DS	**Session**	Graph	**Question Information**	Level 1 Challenge 86

Question description

There are n boys and m girls in a school. Next week a school dance will be organized. A dancing pair consists of a boy and a girl, and there are k potential pairs.

Your task is to find out the maximum number of dance pairs and show how this number can be achieved.

Constraints

- $1 \le n, m \le 500$
- $1 \le k \le 1000$
- $1 \le a \le n$
- $1 \le b \le m$

Input

The first input line has three integers n, m and k: the number of boys, girls, and potential pairs. The boys are numbered 1,2,…,n, and the girls are numbered 1,2,…,m.

After this, there are k lines describing the potential pairs. Each line has two integers a and b: boy a and girl b are willing to dance together.

Output

First print one integer rr: the maximum number of dance pairs. After this, print lines describing the pairs. You can print any valid solution.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 1
3 1
```

EXPECTED OUTPUT

```
2
1 2
2 1
```

Test Case 2

INPUT (STDIN)

```
3 2 4
1 1
1 2
2 3
3 2
```

EXPECTED OUTPUT

```
3
1 1
2 3
3 2
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void link(int u,int v)
```

Test Case 2

KEYWORD

```
int bfs(int n)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
2
```

Test Case 2

TOKEN COUNT

```
600
```

Test Case 3

NLOC

```
84
```